Alexander Debusk

Module 06 Reflection Journal - Regression and Classification
ITAI 1371

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L06 Reflection – Alex DeBusk

In this lab, I learned about the difference between regression and classification and how each one is used for different types of predictions. Regression is used to predict numerical values, while classification is used to predict categories. We used the Titanic dataset to practice both methods, with linear regression predicting passenger fares and logistic regression predicting whether passengers survived. One thing I found interesting was how the same dataset could be used for two different machine learning tasks depending on what we wanted the model to predict.

I was a little surprised at how little there was to do for the actual assignment. I expected there to be more coding involved, but most of the work was simply understanding what each part of the code was trying to do. Once I understood the purpose of the features, targets, training data, and testing data, the rest of the assignment was pretty straightforward. I also learned why we use different evaluation metrics for different models. Mean squared error measures the difference between predicted and actual numerical values, while accuracy measures how often a classification model makes the correct prediction. At first, some of the code looked confusing, but breaking it down into smaller steps made it easier to understand.

Another thing I learned was how important it is to prepare the data before training a model. For example, we had to convert the passenger's sex into numerical values so the model could use it. I also had to fix a small error involving a missing NumPy import, which reminded me that even simple code can have problems if something is missing. These are useful skills that I can use in future machine learning assignments.

Shareefah Adeola

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REFLECTION JOURNAL – Module 06: Regression and Classification

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This lab helped me understand how machine-learning models are trained, tested, and evaluated using the Titanic dataset. I learned that Linear Regression predicts numerical values, such as passenger fares, while Logistic Regression predicts categories, such as whether a passenger survived.

During the lab, I was faced with a syntax error because the code was not properly separated into lines. I also received a NameError because LinearRegression had not been recognized. Correcting the code formatting and rerunning the import cell made me learn the importance of proper Python syntax, imports, and executing cells in the correct order.

My Linear Regression model produced a Mean Squared Error of approximately 3,364.92, while my Logistic Regression model achieved approximately 74.83% accuracy. These results bring to my understanding why MSE measures numerical prediction errors and accuracy measures the proportion of correct classifications.

I also learned that data preparation is essential. The dataset was cleaned, divided into training and testing sets, and prepared for both models. Running the entire notebook successfully reinforced the importance of checking each workflow stage and ensuring that the code runs reproducibly.

Overall, this lab improved my understanding of supervised learning and my confidence in troubleshooting Python errors.

Yilin Leng

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Reflection

Learning about regression and classification is an important step in machine learning and data science. In Lab 06 for ITAI 1371, I used the Titanic dataset in Google Colab to practice these two types of machine learning models. Through this lab, I learned how to train models, make predictions, and check their results.

At first, I found some steps in the machine learning process a little confusing. However, after working through the lab, I understood the two main ideas better. Before this lab, I only knew the basic definitions of regression and classification. After trying both models, I understood their differences more clearly. Regression predicts numerical value, such as a passenger ticket fare. Classification predicts a category, such as whether a passenger survived. Regression focuses on the difference between predicted and actual values, while classification focuses on whether the predicted category is correct.

For the regression task, I used Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) to check the model's predictions. The RMSE was about 58.01, which showed that there was still a gap between the predicted fares and the actual fares. For the classification task, I used Logistic Regression to predict whether passengers would survive. The model reached an accuracy of 74.83%. This helped me understand why different tasks need different evaluation methods.

During the lab, I faced some problems, such as undefined variables and code errors. These problems happened when I ran code cells in the wrong order or when the code was not written in the correct format. I fixed them by checking the code step by step and making sure the

required variables were defined before using them. I also printed the .coef_ attribute of the Linear Regression model. This helped me understand how features such as age and passenger class affect the predicted fare. It made the model easier for me to understand.

This lab helped me learn how to use basic models in the scikit learn library. More importantly, I learned that different machine learning tasks need different models and evaluation methods. We need to choose the right method based on what we want to predict. This experience gave me a better understanding of machine learning and will help me with future studies in deep learning.